Round 14
A different fourth-order closure

One candidate: join the total carried third cumulant to itself through the inverse covariance.

I cannot justify a design that achieves both the leaders' accuracy and their cost. The mechanism below is a genuinely different, inexpensive accuracy experiment. It does not replace the expensive cubic carrier. I would run its falsifier before committing the requested engineering period; I would not describe it as an explanation of the leaders.

Labels: Q = quoted from your prompt or the identified code; D = derived; E = estimated or a proposed design/gate. Equation coefficients and dimensions are D unless identified as design choices. No teacher-data accuracy or grader timing has been measured in this response.

Change the fourth-order closure variable

D. Write C for covariance, K for the symmetric third cumulant, and Y_ia=K_iia. Let Ψ(t) be the log moment-generating function and Γ its local Legendre transform. Differentiating the inverse-Hessian identity gives the exact connected/proper-vertex relation [2]:

$$K^{(4)}_{ijkl} = E_{ijkl} + R_{ijkl},$$

$$E_{ijkl} = K_{ija}(C^{-1})_{ab}K_{klb} + K_{ika}(C^{-1})_{ab}K_{jlb} + K_{ila}(C^{-1})_{ab}K_{jkb}, \qquad R = -C^{\otimes 4}\Gamma^{(4)}.$$

Repeated internal indices are summed. E is the exchange contribution; R is the independent fourth-order vertex contribution. In particular,

$$E_{iiii} = 3Y_{i:}C^{-1}Y^{I}_{i:}.$$

D. For invertible linear B, E(B⊗³K, BCBᵀ)=B⊗⁴E(K,C). Thus E carries the entire cubic history through linear layers without another fourth-order source stack. It has genuine all-distinct entries. It is even under B→−B, as a fourth cumulant must be; the cubic parity obstruction does not rule it out.

What is new, and what is conjectural

D. If K=Σ_b K_b, E contains every cross-birth product K_b C⁻¹ K_c, including b≠c. The rejected Round 13 carrier was a sum of individual-birth quartic forms on P_b. It did not compute these cross-birth, covariance-whitened interactions. Nor is E a covariance-only scale mixture, a new Gaussian birth kernel, or a learned per-neuron correction.

E: the actual hypothesis. On neurons relevant to the mean, the *standardized residual* R_iiii/C_ii² has little pattern beyond a layer-level offset. This is not implied by the exact identity or by your linear-carry measurements. If "κ₄ riding κ₃" already tested this precise inverse-covariance exchange readout, its recorded failure applies and this proposal is closed too.

D. Nothing makes K₄ a function of C and K₃ in general: symmetric distributions can have identical covariance and zero K₃ but different K₄. The proposed approximation is to R, not a proof that independent fourth-order memory is unnecessary.

Concrete estimator and state

D. Retain V29's $\mu \in \mathbb{R}^n$, $C \in \mathbb{R}^{n\times n}$ and complete existing factored K_3 state, including every newborn hub. Retain the regenerated fourth-cumulant channel as a level estimate and for the existing pair closure. No additional source, rank cap or memory horizon is introduced. The new implicit fourth-order object is

$$\widehat{K}^{(4)} = E(K, C) + \rho\, \mathcal{S}_3(C, C),$$

$$[\mathcal{S}_3(C, C)]_{ijkl} = C_{ij}C_{kl} + C_{ik}C_{jl} + C_{il}C_{jk}.$$

E: fixed prototype choices. Query m=384 neurons at pre-activation layers 7-16. Select them by the largest $|\partial \text{mean}/\partial \kappa_4| \cdot \text{var}^2$, computed from the current chain, not the teacher. Keep the baseline κ_4 readout elsewhere. This limits readout work, not history: K still contains every birth. Inactive and always-on birth hubs remain intact.

$$a_i = \frac{(\alpha_i^2 - 1)\phi(\alpha_i)}{24\sigma_i^3}, \quad S = \text{top}_m\{|a_i|\sigma_i^4\}, \quad \alpha_i = \mu_i/\sigma_i.$$

D: implement without an inverse. Put $v_i = C_{ii}$, $R_C = \text{diag}(\sigma)^{-1} C\, \text{diag}(\sigma)^{-1}$, and $Z_{ia} = Y_{ia}/(v_i\, \sigma_a)$, $i \in S$. Solve

$$(R_C + \tau I)X = Z^T, \qquad q_i = 3\sum_a Z_{ia}X_{ai}.$$

E. Pin $\tau = 10^{-3}$ for the first test. This replaces the exact metric in E by a regularized one and sacrifices linear equivariance. Audit stability offline; do not silently change the inverse. Restore $Y_{ii} = d3_i$ if the source routine stores a zero-diagonal D21 slice.

D: level-preserving readout. Let $b_i = \kappa_{4,i}/v_i^2$ be the existing regenerated value. Use no fitted amplitude:

$$c = \frac{1}{m}\sum_{i\in S}(b_i - q_i), \qquad \widehat{k}_{4,i} = v_i^2(q_i + c) \quad (i \in S).$$

This sets $3\rho = c$ and preserves the average standardized level on S. It replaces the baseline pattern there. Do not add it to the full regenerated diagonal, which could double-count the exchange contribution.

Birth, propagation and closure

D. At each ReLU, use V29's source birth and covariance update: $K_{post} = \text{diag}(\Phi) \otimes^3 K_{pre} + B_{V29}$ and $C_{post} = \text{Closure}_{V29}(\mu, C, K, \ldots)$. Transport K and C through the actual W as V29 does. Recompute the implicit E and residual offset at the next queried pre-activation. There is no independent R-memory birth; discarding that memory is the central, falsifiable approximation.

D. Feed $\widehat{\kappa}_4$ into the existing Edgeworth *mean* closure in place of $\widehat{\kappa}_4$. Keep the bivariate closure and its fourth-order slices unchanged initially. This is a hybrid moment estimator, not a claim to represent every fourth-order slice consistently. Changed means feed subsequent layers normally. It is a new fourth-order closure block, not a cheaper cubic architecture.

D/E: working arrays are R_C (n,n), Z (m,n), X (n,m), and LU workspace. Two square buffers plus Z/X require 11 MiB in float32 [D]; selected final-layer source work can raise incremental workspace to about 64 MiB [E]. Existing source memory is unchanged.

FLOP ledger: include the final-layer work V29 skips

Q. n=1024 and 1 u=2^{31} FLOPs. Define G(a,b,c)=ac(2b−1) [D]. The solve has real shapes (1024,1024) and (1024,384). Under the published flopscope rule [3], its bill is floor($2n^3/3+2n^2m$), not the cost of a vector solve.

Increment [D]	Real shape / count [D]	FLOPs or units [D]
One queried layer: solve	n×n system; m right-hand sides	1,521,134,250 FLOPs = 0.708333333 u
Ten queried layers: solves	10 solves; no Strassen calls	7.083333330 u
Final Y: young sources	8 × G(384,1024,1024)	2.998535156 u
Final Y: older tier	6 × G(384,1024,384)	0.843338013 u
Final Y: oldest tier	16 × G(384,1024,224)	1.311859131 u
Nested + shared lifts	G(384,224,384) + G(384,384,1024)	0.171134949 u
Final Y: thin right factors	Formula below; public thin widths	0.546252608 u
Final full covariance	G(1024,1024,1024); reuse C W	0.999511719 u
Major-operation total	Solves + restored output queries	13.953964906 u

D. The output has four young, three rank-384 and eight rank-224 sources. Existing leg transport/re-formation and diagonal-readout buffers are reused. Plain batched products handle the added final queries; no square-Strassen discount is assigned to rectangular shapes.

D, public-code schedule [4]. With k=15 sources, residual width r=18 and feedback width f=16, the extra thin-right-factor cost is

$$k\{G(m,n,r)+G(m,r,n)+2G(m,n,f)+2G(m,f,n)\}+G(m,k,n).$$

Substitute actual widths if your port differs. Final full covariance must also be formed: the baseline's output-only diagonal is insufficient. Omitting that product or the final Y readout materially understates cost.

E: total increment ≈15 u. Allow approximately 1 u for normalization, selected-row gathering, left-factor assembly, reductions and buffer writes beyond the derived 13.954 u. This allowance is estimated, not an exact whole-program meter result. Trace it before submission.

E: residual target. Batch sources by tier, reuse Gaussian coefficients, and add no per-source Python loop. Target ≤300 additional metered calls; estimated network total 17,950-18,000 against the quoted 17,700. This is marginal. A trace above 18,000 or a capped runtime fails the engineering gate; FLOPs do not certify either wall cap.

D: the exchange diagonal and unregularized equivariance were checked on small dense tensors to floating-point roundoff. These are algebra checks, not MLP accuracy or runtime measurements.

One cheap first falsifier: exchange-residual replay

E: one pre-registered experiment, no coefficient fitting. On both stored networks, freeze the V29 trajectory and masks S. Compute the prediction twice at each available queried layer: from teacher C,Y,d3, and from V29 C,Y,d3. Both use the V29 level, fixed τ and fixed coefficient. The teacher calculation is a component oracle, not a deployable result.

```
# S uses V29's sensitivity ranking; m = 384.
# Y includes true/model d3 on its diagonal.
v = C.diagonal()
s = np.sqrt(v)
R = C / s[:, None] / s[None, :]
Z = Y[S, :] / v[S, None] / s[None, :]
X = np.linalg.solve(R + 1e-3*np.eye(n, dtype=R.dtype), Z.T)
q = 3*np.einsum('ia,ai->i', Z, X)
b = k4_v29[S] / (v[S]*v[S])
k4_hat = k4_v29.copy()
k4_hat[S] = v[S]*v[S] * (q + (b-q).mean())
```

D. This NumPy pseudocode is for the stored-array experiment. Deployment must preserve float32, use metered operations and reuse buffers. Teacher and chain covariance-inverse errors may differ sharply; a teacher-only result would conceal that risk.

E: falsification gates. Require the teacher-input predictor to halve the sensitivity-weighted fourth-channel error $\Sigma_i a_i^2(\hat{k}_{4,i} - k_{4,\mathrm{true},i})^2$ relative to V29, pooled over the available deep layers, separately on both networks. Also require centered pattern correlation ≥0.7 on S at every available layer from 12-16. Failure rejects the scalar residual-vertex approximation; do not rescue it with a learned scale.

E. If that passes, the V29-input version must beat V29's same component on both networks. Replay its predicted diagonal in the frozen mean-closure harness, alongside a masked teacher-κ_4 injection. The latter measures this schedule's actual oracle ceiling. If the masked oracle cannot reduce final MSE by 20%, reject this scope for the requested accuracy mechanism. These are parts of one replay, not a parameter sweep.

D/Q. Synthetic teachers cover layers 12, 14, 15 and 16 in the final target range [Q]. Do not report a measured synthetic layer-13 gate. A stored-input replay that passes is still not a free coupled rollout.

D/E: experiment bill. With every requested layer available, teacher float64 solves plus chain float32 solves on both networks cost at most 42.5 u [D] for the solves, plus array/replay work. Stored Y and C avoid final-source reconstruction. Optimized NumPy/BLAS should fit inside a one-hour CPU timebox [E; not timed here].

What the experiment decides

D. It asks whether the missing pattern is exchange memory already encoded in K_3, or independent fourth-vertex memory. Low correlation or a large structured residual answers this before a new carrier, training set or source rewrite is built. The quoted κ_4 oracle alone cannot answer it.

No teacher/source arrays accompanied the prompt. The network falsifier has not been executed. All MLP accuracy claims remain quoted or conditional.

Ceiling, development gates and the remaining gap

Q → D. The full κ_4-diagonal oracle gives a public-network MSE ratio of 0.70. Applied illustratively to V29's quoted 2.13×10^{-8}, that is 1.491×10^{-8} [D]. This is an empirical oracle reference, not a universal lower bound. The selected schedule needs its own oracle: correlated errors prevent assuming monotonic gains. Variance and κ_3 oracle gains cannot be added to it.

Stage [E]	Pass/fail against plain V29 [E]
Days 1-2: stored replay	Pass the fixed falsifier on both networks and establish the masked oracle. Otherwise stop.
Days 3-5: coupled prototype	Use the same public V29 source state. Require better κ_4 component accuracy on both teachers and ≥10% raw-MSE improvement on the 100-network harness.
Days 6-9: engineering	Transfer existing batching, Strassen and role-specific compaction. Require ≤210 u, ≤18,000 traced ops and both hard wall caps on every tested network.
Days 10-14: freeze/validate	Target ≥20% raw-MSE improvement versus V29 with the same formula; confirm held-out. Do not assume untested rank or hub savings.

E. The prototype is approximately 260+15=275 u; transferring your existing engineering gives approximately 193+15=208 u. Neither assumes a new cubic compression. Conditional maturity scenarios, *not performance forecasts*:

Conditional raw result	Score at 208 u [D given E cost]
10% better than V29: 1.917×10^{-8} [D]	3.894×10^{-9}
20% better than V29: 1.704×10^{-8} [D]	3.461×10^{-9}
Full quoted κ_4 oracle: 1.491×10^{-8} [D]	3.029×10^{-9}

D. Even the illustrative full-oracle case misses the quoted leaders' $2.0\text{-}2.1\times10^{-9}$ score. At raw 1.491×10^{-8}, score 2.1×10^{-9} needs cost ≤144.2 u. At 208 u it instead needs raw MSE $\leq1.034\times10^{-8}$. This proposal supplies neither further reduction.

Judgment. The exchange closure warrants the one falsifier because its readout is affordable and distinct from the rejected per-source quartic carrier. Before that test, no accuracy gain is established. I do not have a defensible two-week route to the leaders; their scores alone do not identify their architecture or prove it is outside every public-chain variant.

Sources and provenance

[1] Your attached *ROUND14_prompt_accuracy_mechanism.md*: all quoted teacher, leaderboard and baseline measurements, accepted as supplied rather than reproduced.

[2] S. Floerchinger, Quantum field theory 2, lecture 03: connected four-point / proper-vertex identity. The cumulant specialization and proposed projection here are derived; the source does not claim to solve this MLP problem.

[3] AIcrowd, flopscope cost-model reference, "Linalg direct": solve cost $2n^3/3+2n^2\cdot$nrhs. Checked 27 September 2026.

[4] 504aldo, estimator_v29.py, pinned source: omitted final D21, fused right-factor contractions and thin-factor shapes. Your modified port requires its own trace.